

# Element E&F

STEM Principle Applications  
& Design Viability

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# Problem Statement

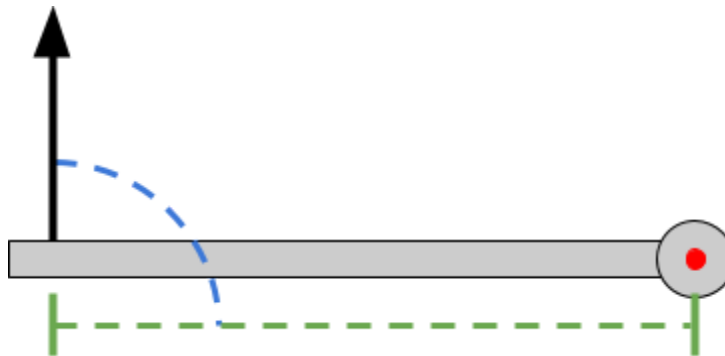
Medical institutions lack a cost-effective, risk-free method for surgical training and other medical processes. Certified cadavers –donations that pass various body aptitude tests– are the best training material for students, but are in great shortage worldwide, posing ethical and practical issues. Due to said scarcity, organizations such as government agencies allocate a great proportion of their budget towards acquiring these materials: Medicare spends ~\$180,000 per medical resident each year through subsidies to universities, totaling over \$25 billion.<sup>1</sup> Overall, inefficient distribution and lack of product availability hinder surgical trainees worldwide, thus a more accessible solution is required.

## I. STEM Principles

### Servo Force Application

A few things must first be calculated to calculate how much force the glove will apply to the fingers. The first of which is the amount of force applied by the strings. This can be calculated by using a rearranged version of the torque equation which solves for force.

$$F = \frac{\tau}{r \sin \theta}$$



**Figure 1.1** Reference Diagram for Torque Calculation. The red dot is the pivot location.

where:

- $\tau$  is Torque in N•M
- $r$  is the distance between the pivot and the force in meters. (Dashed green line in Figure 1.1)
- $f$  is the magnitude of the force in newtons. (Black arrow in Figure 1.1)
- $\theta$  is the angle between the force and the distance to the pivot. (Dashed blue arc in Figure 1.1)

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<sup>1</sup> “The Costs of Training Internal Medicine Residents in the United States.” The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

given that:

- $\tau = 0.1765 \text{ Nm}^2$
- $r = 7.5 \text{ mm}$  or  $0.0075 \text{ m}$
- $\theta = 90 \text{ degrees}$

$$F = \frac{\tau}{r \sin\theta} = \frac{0.1765 \text{ Nm}}{0.0075 \text{ m} \cdot \sin 90} = 23.5 \text{ N}$$

To convert the force applied from the string to the force applied to the opening of the hand the following equation can be used.

$$F_{\text{applied}} = F_{\text{tension}} \cdot \sin\theta$$

where:

- $\theta$  is the angle between the string and the finger.

given:

- $\theta$  is equal to  $18.3 \text{ degrees}$

$$F_{\text{applied}} = F_{\text{tension}} \cdot \sin\theta = 23.5 \text{ N} \cdot \sin(18.3) = 7.26 \text{ N}$$

While  $7.26 \text{ N}$  is nearly as high as higher than the initial  $35 \text{ N}$  target. It is still plenty, the  $35 \text{ N}$  target came from a study which showed that the maximum amount of force that the thumb was able to apply during a variety of different tests was around  $35 \text{ N}$ . However, each finger can apply a different amount of force, with the thumb averaging around  $15 \text{ N}$  of force with a peak of  $35 \text{ N}$ , the pointer finger averaging around  $10 \text{ N}$ , the middle finger, only around  $5 \text{ N}$ , and similar for the ring finger. While data was not given for the pinky it can be inferred to be around the same or less than that of the ring finger. So although  $7.26 \text{ N}$  is not enough to completely overpower every single finger, it is still enough for most fingers and enough for a majority of situations.<sup>3</sup>

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<sup>2</sup> Cytron Marketplace. (n.d.). MG90S Metal Gear Micro Servo. Cytron. Retrieved 2 23, 2025, from <https://www.cytron.io/p-mg90s-metal-gear-micro-servo>

<sup>3</sup> Riddle, Michael et al. (2020). Evaluation of individual finger forces during activities of daily living in healthy individuals and those with hand arthritis. Journal of Hand Therapy, 33(2), 188-197. [https://www.jhandtherapy.org/article/S0894-1130\(20\)30079-X/abstract](https://www.jhandtherapy.org/article/S0894-1130(20)30079-X/abstract)

## Max Voltage and Amperage

The final product uses both servos and potentiometers in its operation. To ensure the product works as it should, the amperage draw needs to be less than the amperage supply. The 5v sources are the two USB-C batteries, which each supply 5v at 2 Amps.

| Five Volt Sources    |         |                 |                   |
|----------------------|---------|-----------------|-------------------|
| Device               | Voltage | Max Discharge   | Amp-hour          |
| Battery 1            | 5V      | 2000mA          | 5000 mAh          |
| Battery 2            | 5V      | 2000mA          | 5000 mAh          |
| Parrallel Connection | Same    | 2000mA + 2000mA | 5000mAh + 5000mAh |
| Total                | 5V      | 4000mA          | 10000mAh          |

**Figure 2.1:** Table showing the amperage supply for the 5V circuit with calculations for total maximum discharge and amp-hours.

When run in parallel, the resulting batteries supply 5v at 4000mA with 10000mAh. The 5v sinks are the microcontroller, which draws 280mA<sup>4</sup>, and the 5 servos, which each draw a varying amount of voltage depending on the situation.<sup>5 6</sup>

| Five Volt Sinks       |         |                   |                  |                   |                   |                   |                   |
|-----------------------|---------|-------------------|------------------|-------------------|-------------------|-------------------|-------------------|
| Device                | Voltage | Idle              | No Load          | Light load        | Typical Load      | Heavy Load        | Stall             |
| Microcontroller       | 5v      | 280mA             | 280mA            | 280mA             | 280mA             | 280mA             | 280mA             |
| 5x Servos             | 5v      | 2.7mA             | 70mA             | 150mA             | 225mA             | 300mA             | 700mA             |
| Parrallel Connections | Same    | 280mA + 2.7mA * 5 | 280mA + 70mA * 5 | 280mA + 150mA * 5 | 280mA + 225mA * 5 | 280mA + 300mA * 5 | 280mA + 700mA * 5 |
| Total                 | 5v      | 293.5mA           | 630mA            | 1030mA            | 1405mA            | 1780mA            | 3780mA            |

**Figure 2.2:** Table showing the amperages of the servos at different loads with calculations for total amperage draw.

When run in parallel the resulting amperage draw can range between 293.5mA and 3780mA. This falls below the maximum 4000mA supply of the batteries, so there should be no problems with

<sup>4</sup> Arduino. (n.d.). *Arduino Nano ESP32*. Arduino Documentation. Retrieved February 20, 2025, from <https://docs.arduino.cc/resources/datasheets/ABX00083-datasheet.pdf>

<sup>5</sup> MakerPortal. (n.d.). *MG90S Micro Servo*. Maker Portal. Retrieved February 23, 2024, from <https://makersportal.com/shop/mg90s-micro-servo?srsId=AfmBOornIVRfhN8kAl5EfT-hBz3-akySV6iWfo4ml5M1fll8BdZRDvJ5>

<sup>6</sup> Proto Supplies. (n.d.). *Servo Motor Micro MG90S*. Proto Supplies. Retrieved February 23, 2024, from <https://protosupplies.com/product/servo-motor-micro-mg90s/>

the 5v circuit of the glove. The only 3.3V source in the glove is the microcontroller, which supplies 200mA.

| Three Volt Sources |         |               |          |
|--------------------|---------|---------------|----------|
| Device             | Voltage | Max Discharge | Amp hour |
| Micro Controller   | 3.3V    | 200mA         | 0mAh     |
| Total              | 3.3V    | 200mA         | 0mAh     |

**Figure 2.3:** Table showing the amperage supply for the 3.3V circuit.

The only 3.3V sinks in the glove are the five potentiometers which each draw ~15mA.

| Three Volt Sinks      |         |          |
|-----------------------|---------|----------|
| Device                | Voltage | Usage    |
| Potentiometers x5     | 3.3v    | 15mA     |
| Parrallel Connections | Same    | 15mA * 5 |
| Total                 | 3.3V    | 75mA     |

**Figure 2.4:** Table showing the amperage draw for the 3.3V circuit with calculations for total demand.

In total the 3.3V circuit requires 75mA. This is less than half of the 200mA supply from the microcontroller, so there also should be no problems with the amperage of the 3.3V supply. Because the 3.3V circuit is created by the microcontroller, which is powered by the 5V circuitry, the only amperage which needs to be included for the battery life calculations is from the 5.5V circuit.

| Battery Life          |                    |                  |                   |                   |                   |                   |
|-----------------------|--------------------|------------------|-------------------|-------------------|-------------------|-------------------|
| Device                | Idle               | No Load          | Light load        | Typical Load      | Heavy Load        | Stall             |
| Amperage              | 293.5mA            | 630mA            | 1030mA            | 1405mA            | 1780mA            | 3780mA            |
| Total Amp-Hours       | 10000 mAh          | 10000 mAh        | 10000 mAh         | 10000 mAh         | 10000 mAh         | 10000 mAh         |
|                       | 10000mAh / 293.5mA | 10000mAh / 630mA | 10000mAh / 1030mA | 10000mAh / 1405mA | 10000mAh / 1780mA | 10000mAh / 3780mA |
| Hours of Battery Life | 34.07 Hours        | 15.87 Hours      | 9.71 Hours        | 7.12 Hours        | 5.62 Hours        | 2.65 Hours        |

**Figure 2.5:** Table showing the battery life of the glove given different loads on the servos with calculations.

Depending on the intensity of the load on the servos the glove has varying battery lifes. If the servos simply maintain their positions without load, the glove will last nearly 16 hours. Even if the servos are constantly at the maximum load possible, the glove will still last for 2.5 hours. Which is longer than the battery life of the Meta Quest headset it will be used with.

## Material Comparison

Due to the product having many moving parts, using proper materials is necessary to ensure the product works as well as possible and for as long as possible. The Glove requires a mixture of both rigid and flexible materials, so different tables have been made to compare them. Rigid Materials have been separated into two categories: Plastics, and Metals and Exotics. This has been done because it is much easier to compare similar materials.

When comparing rigid materials for usage in a VR Glove, there are three important qualities: price, density, and yield strength. While other factors will be included in the comparison they are less important.

| Plastics <sup>7 8</sup> |                                                                                                                                                                                            |                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Material                | Pros                                                                                                                                                                                       | Cons                                                                           |
| <b>PLA</b>              | Low-cost . . . . . 10 - 15\$ per KG Spool<br>Lightweight . . . . . 1.24 g/cm <sup>3</sup><br>High Yield Strength . . . . . ~40 MPa<br>Readily Available<br>Easy to 3D Print                | Low Melting Temperature<br>Not Biodegradable                                   |
| <b>PETG</b>             | Low-cost . . . . . 25\$ per KG spool<br>Lightweight . . . . . 1.27 g/cm <sup>3</sup><br>High Yield Strength . . . . . 53 MPa<br>Readily Available<br>Higher Melting Temperature<br>Durable | Harder to 3D Print                                                             |
| <b>ABS</b>              | Lightweight . . . . . 1.04 g/cm <sup>3</sup><br>Easy to Print<br>Impact Resistant                                                                                                          | Expensive . . . . . 30\$ per KG spool<br>Low Yield Strength . . . . . 27.5 MPa |
| <b>Polycarbonate</b>    | Lightweight . . . . . 1.2 g/cm <sup>3</sup><br>High Yield Strength . . . . . 65.5 MPa<br>It turns yellow over time                                                                         | Expensive . . . . . 40\$ per KG spool                                          |

<sup>7</sup> Curbell Plastics. (n.d.). Plastic Properties Table. Curbell Plastics. Retrieved February 23, 24, from [https://www.curbellplastics.com/resource-library/material-selection-tools/plastic-properties-table/?srsltid=AfmBOor6oxLTk\\_5kGargJHvNwvAa4W4Kct4sRNninqw76qbBJknRzUeR](https://www.curbellplastics.com/resource-library/material-selection-tools/plastic-properties-table/?srsltid=AfmBOor6oxLTk_5kGargJHvNwvAa4W4Kct4sRNninqw76qbBJknRzUeR)

<sup>8</sup> Simplify3D. (n.d.). Filament Properties Table. Simplify3D. Retrieved 2 23, 2024, from <https://www.simplify3d.com/resources/materials-guide/properties-table/>

Based on the table above, the best rigid plastic material to go with would be PETG, polyethylene terephthalate glycol, as it excels in all three of the main categories. PETG is a fairly lightweight material that is low-cost, easy to acquire, and relatively strong. Although it can be harder to 3D print, the increase in strength, durability, and melting temperature easily makes up for it.

| Metals and Exotics <sup>9</sup>          |                                                                                                                                                                     |                                                                                                |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Material                                 | Pros                                                                                                                                                                | Cons                                                                                           |
| <b>Aluminum</b> <sup>10</sup>            | Lightweight . . . . . 2.7g/cm <sup>3</sup><br>High Yield Strength . . . . . 193 MPa<br>High Tensile Strength . . . . . 227 MPa<br>Corrosion Resistant               | Expensive . . . . . 2.67\$ per KG<br>Machining Costs                                           |
| <b>Titanium</b> <sup>11</sup>            | High Yield Strength . . . . . 255 MPa<br>High Tensile Strength . . . . . 434 MPa<br>High Strength-to-Weight Ratio                                                   | Expensive . . . . . 17\$ per KG<br>Machining Costs<br>Heavy . . . . . 4.5 g/cm <sup>3</sup>    |
| <b>Stainless Steel 304</b> <sup>12</sup> | High Yield Strength . . . . . 193 MPa<br>High Tensile Strength . . . . . 620 MPa<br>Corrosion Resistance                                                            | Expensive . . . . . 4\$ per KG<br>High Machining Costs<br>Heavy . . . . . 7.8g/cm <sup>3</sup> |
| <b>Carbon Fiber</b> <sup>13</sup>        | High Yield Strength . . . . . 4140 MPa<br>High Tensile Strength . . . . . 3380 MPa<br>Light Weight . . . . . 1.6 g/cm <sup>3</sup><br>High Strength-to-Weight Ratio | Expensive Manufacturing Process                                                                |

Although metals and carbon fiber have use cases where they are highly effective. For the use case of a VR glove, the increase in yield Strength and resistance to elements does not justify the increase in manufacturing costs or the increase in weight. So for this project metals and similar materials will not be used.

<sup>9</sup> Sam. (2022, March 13). *Metal Strength Chart: A Detailed Guide to Metal Strengths*. AT-Machining. Retrieved February 23, 2025, from <https://at-machining.com/metal-strength-chart/>

<sup>10</sup> Daily Metal Prices. (n.d.). *Metal Spot Price Charts*. Daily Metal Prices. Retrieved February 24, 2024, from <https://www.dailymetalprice.com/metalpricecharts.php?c=al&u=kg&d=0>

<sup>11</sup> Shanghai Metal Market. (n.d.). *Titanium Prices*. Metal. Retrieved February 23, 2024, from <https://www.metal.com/en/markets/23>

<sup>12</sup> Kalp Industries. (n.d.). *Stainless Steel price ₹ per kg | SS 304, 316 and 310 Rate*. Kalp Industries. Retrieved February 23, 2025, from <https://www.kalpind.com/stainless-steel-price-per-kg/>

<sup>13</sup> MatWeb. (n.d.). *Overview of materials for Epoxy/Carbon Fiber Composite*. MatWeb Material Property Data. Retrieved February 23, 2024, from <https://www.matweb.com/search/datasheet.aspx?matguid=39e40851fc164b6c9bda29d798bf3726&ckck=1>

When comparing flexible materials for usage in a VR Glove, there are three important qualities: price, density, and shore hardness. While other factors will be included they are less important.

| Flexible Plastics                 |                                                                                                                          |                                                           |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Material                          | Pros                                                                                                                     | Cons                                                      |
| <b>Ninjaflex TPU<sup>14</sup></b> | Lightweight . . . . . 1.2 g/cm <sup>3</sup><br>Shores A Hardness . . . . . 85<br>Easy to 3D Print<br>Abrasion Resistance | Expensive . . . . . 100\$ per KG spool                    |
| <b>Bambo Lab TPU<sup>15</sup></b> | Lightweight . . . . . 1.2 g/cm <sup>3</sup><br>Shore A Hardness . . . . . 95<br>Low-cost . . . . . 49\$ per KG Spool     | Hard to 3D Print                                          |
| <b>Nylon</b>                      | Lightweight . . . . . 1.14 g/cm <sup>3</sup><br>Shore A Hardness . . . . . 70                                            | Expensive . . . . . 80\$ per KG spool<br>Hard to 3D Print |

Based on the table above, the best flexible material for this application is Bambo Lab’s TPU. This is because although it is a harder material, having a shore rating of 95A instead of 85A or 70A, the price per KG is much lower.

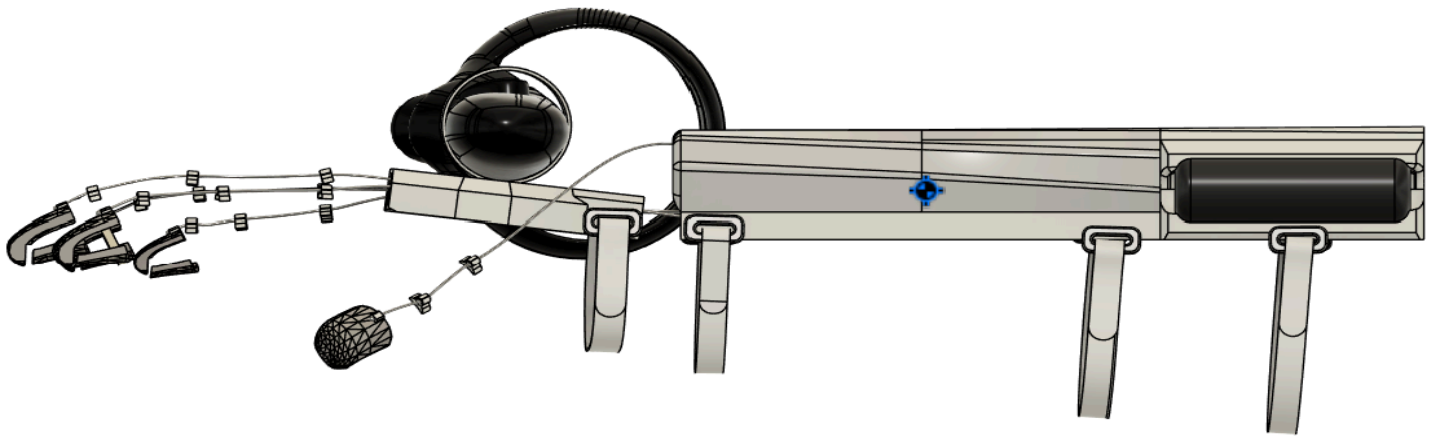
Although these tables do show that the final product should use PETG and Bambo Lab’s TPU, the prototype will be made of PLA and Ninjaflex TPU as those parts are locally available which helps to reduce the price of the prototype.

<sup>14</sup> NinjaTek. (n.d.). *NinjaFlex 3D Printer Filament (85A)*. NinjaTek. Retrieved February 23, 2025, from <https://ninjatek.com/shop/ninjaflex/#tech-specs>

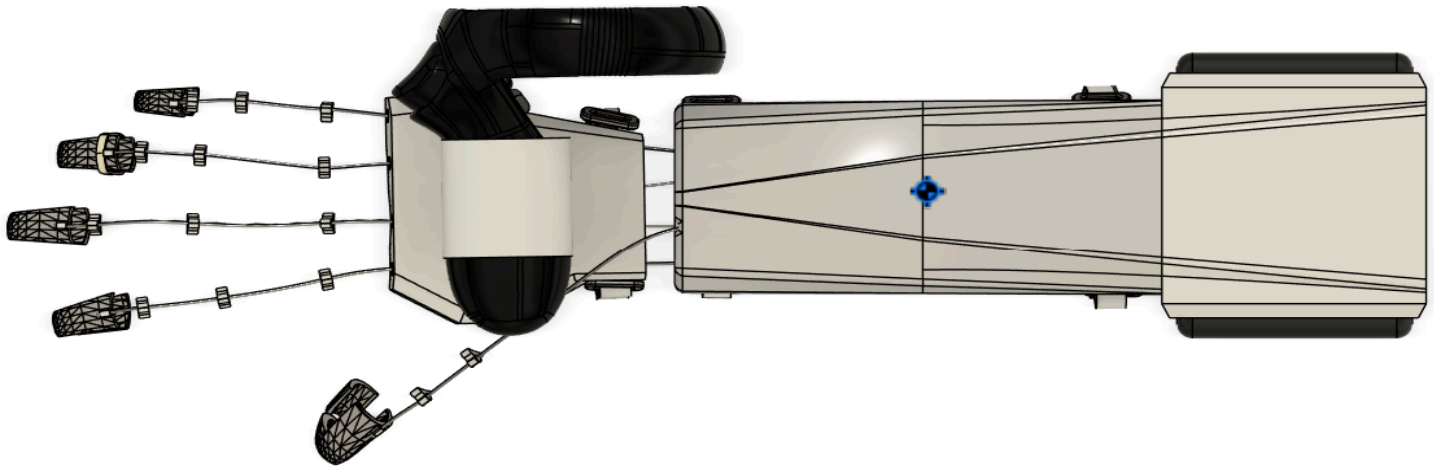
<sup>15</sup> Bambu Lab. (n.d.). *TPU 95A HF*. Bambu Lab. Retrieved February 23, 2024, from [https://us.store.bambulab.com/products/tpu-95a-hf?srsId=AfmBOorZZ78sTbns7Ysmfn\\_00wbMhF5V\\_huY8sx5cc0KSSS6Trn7fL1O](https://us.store.bambulab.com/products/tpu-95a-hf?srsId=AfmBOorZZ78sTbns7Ysmfn_00wbMhF5V_huY8sx5cc0KSSS6Trn7fL1O)

## Weight and Center of Mass Comparison

The weight of the glove can make or break the experience for the user. To create an estimated weight for the glove, each component in the CAD file was given materials with the correct density of the parts in real life and was then used to calculate the total weight of the glove, which came out to 750g, higher than the 400g maximum weight product specification that was the initial target. However, this is still acceptable as the center of mass has been moved to make the weight seem lighter. The center of mass can also be calculated through the CAD model, which gives the result that the center of mass of the glove is located 295mm from the tip of the middle finger.



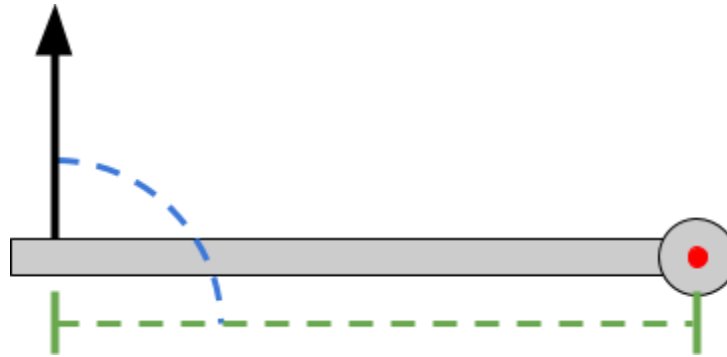
**Figure 3.1** Front view of the glove with center of mass included.



**Figure 3.2** Side view of the glove with center of mass included.

To compare the effect of the weight of the final solution to other gloves the torque that the glove applies to the arm needs to be calculated. Then using that torque, calculations for how weight would apply the same torque to the arm at the center of mass of other products.

$$\tau = r F \sin\theta$$



**Figure 3.3** Reference Diagram for Torque Calculation. The red dot is the pivot location.

where:

- $\tau$  is Torque in  $\text{N}\cdot\text{M}$
- $r$  is the distance between the pivot and the force in meters. (Dashed green line in Figure 3.3)
- $f$  is the magnitude of the force in newtons. (Black arrow in Figure 3.3)
- $\theta$  is the angle between the force and the distance to the pivot. (Dashed blue arc in Figure 3.3)

Using values obtained from Fusion 360, and the average length of various body parts, the torque from the glove can be calculated.

$$r = D_{\text{CM} - \text{Elbow}} = D_{\text{ToF} - \text{Elbow}} - D_{\text{ToF} - \text{CM}}$$

where:

- $D_{\text{CM} - \text{Elbow}}$  is the distance from the center of Mass to the elbow
- $D_{\text{ToF} - \text{Elbow}}$  is the distance from the tip of the middle finger to the elbow
- $D_{\text{ToF} - \text{CM}}$  is the distance from the tip of the middle finger to the Center of Mass.

Given that:

- $D_{\text{ToF} - \text{Elbow}}$  is equal to  $475 \text{ mm}^{16}$
- $D_{\text{ToF} - \text{CM}}$  is equal to  $295 \text{ mm}^{17}$

$$r = 475\text{mm} - 295\text{mm} = 180 \text{ mm} = 0.18 \text{ m}$$

$$F = ma$$

where:

- $F$  is the magnitude of the force
- $m$  is the mass of the object in KG
- $a$  is the acceleration

<sup>16</sup> City University of Hong Kong. (n.d.). 23. *Elbow-fingertip length*. Anthropometrics. Retrieved February 23, 2024, from <https://personal.cityu.edu.hk/meachan/online%20anthropometry/Chapter2/Ch2-23.htm>

<sup>17</sup> City University of Hong Kong. (n.d.). 28. *Hand length*. Anthropometrics. Retrieved February 23, 2024, from <https://personal.cityu.edu.hk/meachan/online%20anthropometry/Chapter2/Ch2-28.htm>

given:

- $a = 9.8 \frac{m}{s^2}$  on Earth
- $m = 750g$  or  $0.75kg$

$$F = 9.8 \frac{m}{s^2} \cdot 0.75kg = 7.45 N$$

Assuming  $\theta$  to be equal to 90 degrees, and by plugin the values above into the initial torque equation the following is given:

$$\tau = r F \sin\theta = 0.18 m \cdot 7.45 N \cdot \sin(90 \text{ deg}) = 1.341 Nm$$

With this torque value, a new force can be calculated given that the force is at the center of mass of other products.

$$r = D_{\text{MoH} - \text{Elbow}} = D_{\text{ToF} - \text{Elbow}} - D_{\text{ToF} - \text{MoH}}$$

where:

- $D_{\text{CM2} - \text{Elbow}}$  is the distance from the center of Mass to the elbow
- $D_{\text{ToF} - \text{Elbow}}$  is the distance from the tip of the middle finger to the elbow
- $D_{\text{ToF} - \text{MoH}}$  is the distance from the tip of the middle finger to the middle of the hand.

given:

- $D_{\text{ToF} - \text{Elbow}} = 475 \text{ mm}$
- $D_{\text{ToF} - \text{MoH}} = 91 \text{ mm}^{18}$

$$r = 475mm - 91mm = 384 \text{ mm} = 0.384 m$$

Plugging that into a rearranged torque equation you get the following:

$$F = \frac{\tau}{r \sin\theta} = \frac{1.341 Nm}{0.384 m \cdot \sin(90 \text{ deg})} = 3.492 N$$

Finally, the equivalent weight can be calculated by plugging that value into a rearranged force equation and solving for the mass.

$$m = \frac{F}{a} = \frac{3.492 N}{9.8 \frac{m}{s^2}} = 0.356 kg = 356 g$$

The equivalent weight for this glove is 356 grams which is less than the maximum weight previously decided upon.

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<sup>18</sup>City University of Hong Kong. (n.d.). 28. *Hand length*. Anthropometrics. Retrieved February 23, 2024, from <https://personal.cityu.edu.hk/meachan/online%20anthropometry/Chapter2/Ch2-28.htm>

## Digital Displacement & Rotation

Much of the product is the Test Application (TA) used with the glove. This prototype would use the TA as an open-source base for developers to create training simulations, viable with current medical applications. A key aspect of the TA would be a sense of spatial understanding for immersion within the digital world, akin to the real world.

Unity will be used as the game engine to design the TA for its expansive VR compatibility. As such, the native VR controller should automatically be tracked in the base TA. To improve spatial accuracy, established scales (inches and degrees) will be in place to simulate real distances. These give a good foundation to the tween digital movement. A primitive conversion factor of tracked real displacement ( $x_r$ ) will be translated to the digital hand's displacement ( $x_d$ ) following this equation:

$$x_d = ax_r^n$$

where:

- $x_d$  is the digital displacement
- $x_r$  is the real displacement
- $a$  is the conversion coefficient
- $n$  is the linearizing degree

Similarly, for rotational displacement  $\theta_r$ , the conversion equation is:

$$\theta_d = b\theta_r^m$$

where:

- $\theta_d$  is the digital rotational displacement
- $\theta_r$  is the real rotational displacement
- $b$  is the conversion coefficient
- $m$  is the linearizing degree

The conversion coefficient and linearizing degree will ensure both displacements are proportional to themselves and the established scales.

As for adjusting the values during the testing phase, we may utilize a t-distribution to collect sample positions and rotations, crafting a sampling distribution to estimate the true mean difference between real and digital data.

## Forward Kinematics and Digital Animation

Within the glove, servo motors control the movement of each finger (as stated in the Servo Force Application section), translating rotational servo movement to finger extension. To accurately

reflect this motion in a rigged 3D hand model, a forward kinematics (FK) system is implemented, a basic concept in animation and 3D modeling.

FK determines the end position of a jointed system given a set of rotation inputs. Servo motor rotation angles would directly affect the bending angle of the corresponding digital finger joints.

For each finger joint rotation, FK follows this equation:

$$P_{n+1} = P_n + R(\theta_n) \cdot L_n$$

where:

- $P_n$  is the position of the current joint
- $R(\theta_n)$  is the rotation angle applied to the joint based on servo rotation  $\theta_n$
- $L_n$  is the length of the phalanx segment

This equation is simplified for conceptual understanding, but the basic process is iterative, where the angle of each segment follows the one before it, similar to how our real finger joints work. Each servo reports its current rotational angle ( $\theta_n$ ) via a potentiometer, which allows real-time updates of the digital hand model's finger positions.

These equations are handled in Unity using their Humanoid Rig System<sup>19</sup>, where finger 'bones' are mapped to corresponding servo-driven finger joints and all calculations are done to process those values and apply them to the model using FK. Additionally, Unity's Transform Hierarchy<sup>20</sup> allows each phalanx to be rotated independently, ensuring follow-through articulation of the fingers.

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<sup>19</sup> Unity. (2015). Unity Documentation. <https://docs.unity3d.com/510/Documentation/Manual/UsingHumanoidChars.html>

<sup>20</sup> Unity. (2015). Unity Documentation. <https://docs.unity3d.com/Manual/class-Transform.html>

## VR Application Inclusion

The Test Application (TA) that is paired with the glove is extremely important not only to test the glove but also to serve as a base for potential medical applications in the future. The product hopes to be a standard for developers and established medical applications could be integrated with it. This section will go in greater detail.

To create the TA, Unity will be used as the main game engine due to its native VR support (called VRCORE) and compatibility with SteamVR's SDK, an established VR platform for integrated headsets, controllers, and additional components into these applications. A SteamVR Unity plugin would allow handling input, loading models, and processing basic collision actions<sup>21</sup>.

VR locomotion, grabbing, and feedback functions are built-in, reducing custom development time. Delving deeper into SteamVR, the base within Unity's VRCORE platform, it provides:

### 1. Hand Tracking

The Oculus controller's tracking would be linked to SteamVR, providing the hand position & rotation data required whilst controlling the other features/functionalities of the controller. This would also reduce hardware complexity as the product would not require its own sensors. Additionally, the SteamVR Input System would allow for easy remapping of real positions to a digital hand model. In fact, SteamVR natively comes with a basic hand model that we can adjust to fit the glove, with built-in finger curl detection that can be used (see Forward Kinematics and Digital Animation).

### 2. Collision Detection & Force Feedback Procession

SteamVR has prebuilt basic physics-based interactions (grabbing, throwing, touching objects). Additionally, Interactable Components and Unity's prebuilt `OnCollisionEnter()` functions allow for more customization with glove collision handling. This means it should be easy to integrate with any custom code.

### 3. Expandability

SteamVR is an established platform that works with Unity, is constantly updated, and is compatible with most mainstream VR headset lines, not just Oculus.

Overall, Unity has its base components, and SteamVR can help with additional aspects of the TA, but custom code will also be required to complete it.

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<sup>21</sup> Bradner, K., Leiby, A., Selan, J., Aleiby, & Brown, N. (2024, January 15). SteamVR Unity Plugin GitHub. [https://github.com/ValveSoftware/steamvr\\_unity\\_plugin?tab=readme-ov-file](https://github.com/ValveSoftware/steamvr_unity_plugin?tab=readme-ov-file)

Here is a breakdown of all the components, split between SteamVR/Unity's base and custom code.

| Component                            | SteamVR Development                     | Custom Development                                                                          |
|--------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------|
| Hand Tracking (Position & Rotation)  | Yes, basic component within VR platform | Yes, custom code to shift proportions (see Digital Displacement & Rotation)                 |
| Finger Tracking                      | No, only detects the controller grip    | Yes, use servo data + FK in Unity Humanoid Rig System                                       |
| Collision Detection                  | Yes, SteamVR physics and colliders      | No additional work is needed, might need additional code to fit within medical applications |
| Force Feedback (Restricting Fingers) | No                                      | Yes, custom force calculations + servo control from SteamVR collision data                  |
| Wireless/Bluetooth Communication     | No                                      | Yes, manual Unity-to-Arduino pipeline required                                              |
| Medical Application                  | No                                      | Yes, needs custom code to build simulations                                                 |

## II. Laws and Regulations

Not many current laws dictate VR extension products. The only laws that would directly affect the glove are related to the batteries and the wireless communication from the microcontroller. Neither of these should be problems as the batteries are sold on Amazon, which requires battery packs to have a UN 38.3 lithium battery test summary for the product to be sold.<sup>22</sup> The microcontroller also lists certificates to prove that it abides by FCC regulations.<sup>23</sup>

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<sup>22</sup>Mallocci, I. (2024, May 9). *Amazon Lithium Batteries Requirements For Sellers: An Overview*. Compliance Gate. Retrieved February 23, 2024, from <https://www.compliancegate.com/amazon-lithium-batteries-requirements/>

<sup>23</sup>Arduino. (n.d.). *Product Compliance*. Arduino Docs. Retrieved February 23, 2024, from <https://docs.arduino.cc/certifications/?search=nano+esp>

# III. Development

The development and production of the haptic VR glove will primarily focus on eco-friendly materials and processes to minimize harmful environmental impacts. Careful refinement, efficient production and operation, and recycling efforts during the end-of-life phase can mitigate some of these effects while maintaining a strong target market.

## Premanufacturing

### 1. Raw Material Extraction

The haptic glove primarily uses NinjaFlex TPU for the 3D prints, which is a thermoplastic polyurethane that is considered highly safe and is resistant to chemicals, abrasion, and outdoor elements. It is also recyclable, making it an eco-friendly option. The spandex fabric that is used in the glove is created from mainly polyurethane in addition to nylon and polyester. The electronics used consist of copper, silicon, and rare earth metals, and the mining processes can be harmful due to the high energy consumption and pollution from extraction.

### 2. Material Refinement

The polyurethane that is created to form TPU is chemically processed, but all the chemicals used are non-toxic and safe based on food standards. The spandex, however, involves toxic solvents and high water usage to create its stretch and recovery properties, as well as its heavy reliance on synthetic dyes. This causes it to become non-biodegradable, harming the environment in the long-run. Some of the electronics components are refined and purified, which creates toxic waste, but careful disposal and recycling efforts can be made to prevent environmental damage.

## Manufacturing

### 1. Component Manufacturing

The glove's structural components(servo mounts, string tensioners, electronics holders) are printed with NinjaFlex TPU, the electronics(Arduino, motors, potentiometers, batteries) are produced and assembled separately before being incorporated into the glove, and the glove itself is likely machine sewn for mass production. This section of the product life cycle will likely remain ineffective towards the environment and society.

### 2. Assembly

The 3D-printed components are smoothed and sanded before being joined together and integrated with the electronic parts, the fabric and velcro straps are fitted on to ensure a secure fit for all hands sizes, and the entire system is tested for calibration and durability before packaging.

Dependence on multiple suppliers for various materials may create production delays, which can hinder product marketability in society.

## Product Delivery

### 1. Packaging

Packaging is easy to manage in terms of impact. Biodegradable or recycled boxing with thick cardboard would be much better for the environment and allow for a more premium package like other high-end tech products. Since the product is modular, a minimalist design would reduce material waste by packing each component strategically.

### 2. Distribution

The main target market is medical institutions that would buy in bulk without constant shipments, and as such there isn't a huge need for quick delivery times; these products won't necessarily be shipped to individual customers as much. Thus, a more sustainable distribution method such as sea freight shipping<sup>24</sup> would be acceptable, as air freight produces 47x more CO<sub>2</sub> per ton-mile than sea freight. Opting for low-emission transport would mitigate impact. If the company is greatly established, regional warehouses would be even more sustainable, reducing delivery distances and both improving times and environmental impact. It is extremely hard to reduce distribution emissions, but by opting for better choices (sea over air) the company would be mindful of their ecological influence.

## Usage

### 1. Installation

The glove is plug-and-play with existing VR systems, requiring minimal setup from a customer. Wired or wireless use, along with hot-swappable batteries, allows for ease of use in any situation.

### 2. Usage

The main environmental impact of the product at this stage is the power required for the product and its connected VR systems to function. These servos and microcontrollers would consume energy over time. Components can gain this energy from electricity generated from various sources, from coal and natural gas to solar and wind, depending on the location of the user and where one would get their energy from. Additionally, a hot-swappable battery would probably increase energy usage as the glove allows for longer VR sessions. However, rechargeable batteries would reduce battery waste as opposed to disposable alternatives and would be more efficient with electricity usage. These modular battery replacements would also increase the lifespan of the glove itself.

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<sup>24</sup> Baxter, K. (2022, April 6). Air Freight vs Ocean Freight Carbon Footprint & Environmental Impact. InTek Freight & Logistics. Retrieved February 23, 2025, from <https://www.inteklogistics.com/blog/air-freight-vs-ocean-freight-carbon-footprint-environmental-impact>

## End of Life

### 1. Disposal

The glove would need to be disposed of if any part of it begins to repair, whether it be the servos losing grip, strings breaking, finger caps and bases being worn out, or if the microcontroller is outdated from the software's code. Although some of these components are fixable, degradation is unstoppable.

The various plastic parts of the glove are non-biodegradable and would require specialized recycling at specific plants for other products or refurbished into new gloves. As for electronics, e-waste is a growing, prevalent problem. Batteries would pose fire hazards if not properly disposed of, and microcontrollers would contain metals, contaminating soil and water with toxic minerals. Although a bit more expensive, instating a return-and-recycle policy similar to other tech companies (such as Apple) would allow reusing these chips and their materials to manufacture more products<sup>25</sup>. Partnerships with recycling facilities and chip manufacturers would allow for less e-waste, for a more sustainable product.

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<sup>25</sup> Apple. (2024). *2024 Environmental Progress Report*. Apple. Retrieved February 23, 2025, from [https://www.apple.com/environment/pdf/Apple\\_Environmental\\_Progress\\_Report\\_2024.pdf](https://www.apple.com/environment/pdf/Apple_Environmental_Progress_Report_2024.pdf)

# IV. Market

## Market Competition

The current market for haptic VR gloves includes major competitors such as SenseGlove, Haply, TouchDIVER, Contact CI, and several others. These companies offer various features and feedback options to their gloves such as exotendon-based gloves, microfluidic actuators, and complex positional tracking. Additionally, some gloves utilize pneumatic air-actuated systems that create pressure sensations on the skin, while others have vibrational elements to simulate force feedback. Furthermore, augmented and mixed reality VR solutions offer more immersive experiences, competing indirectly by providing alternative training environments. Traditional cadaver-based training and real-life practice also remain strong competitors, as they continue to be a reliable way for medical training despite their high costs.

Our solution competes primarily on cost-effectiveness, weight, and a one-size-fits-all design, making it accessible to a broad range of users. Unlike existing gloves that only come in one fitting, our glove is designed for adaptability, allowing medical students to practice surgical techniques in their dorms or with study groups. Additionally, it offers long battery life and repairability without expensive parts, ensuring reliability and affordability. Compared to current VR training solutions, our glove provides an immersive and adaptive experience while remaining significantly more budget-friendly. Future competition could come from large tech companies like Meta and Apple or specialized businesses focusing on haptic VR, though barriers such as high manufacturing costs, existing patents, and lack of technical expertise may slow their entry into the market. With existing haptic VR gloves ranging from \$3,750 (Contact CI VR) to \$6,000 (SenseGlove Nova 2), our solution presents a more cost-effective alternative, making VR-based surgical training more accessible. Furthermore, beyond medical training, it has potential applications in training AI-driven surgical robots faster, remote patient analysis, and building muscle memory for any tasks.

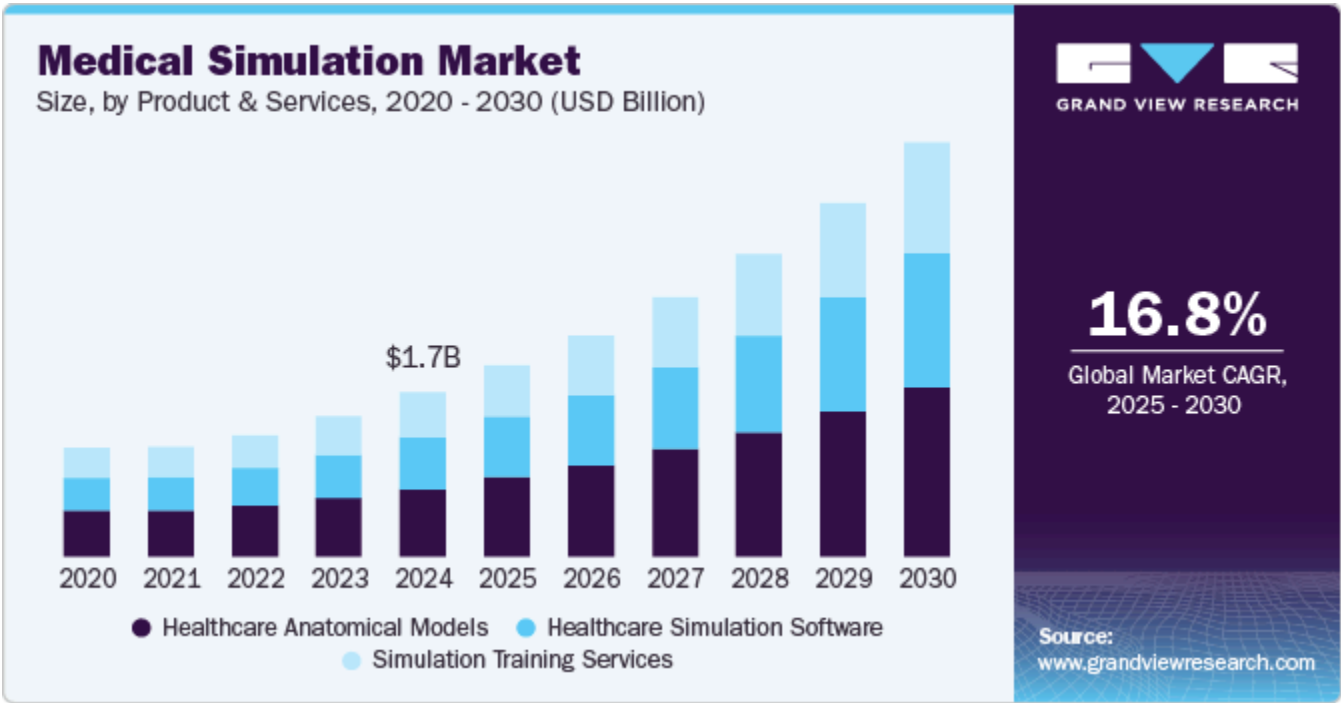
## Users and Buyers

Our primary users for our haptic glove are medical students and trainees that can use our cost-effective and accessible product to enhance medical training in practice procedures in a controlled environment. Unlike traditional training methods, our haptic VR glove will allow users to repeatedly perform surgical techniques while adjusting the difficulty level to progress their skills. This adaptability makes it ideal for refining precision, ensuring students gain hands-on experience without the limitations of cadaver shortages or expensive physical simulators.

The primary buyers of our solution would be medical schools and universities seeking to enhance their simulation labs but could be bought by VR companies interested in licensing our software or incorporating components of our glove into their existing platforms. Our gloves can be modified to perfect any technique in a risk-free environment, and by joining with existing VR training solutions, our product can make hands-on practice more efficient and accessible.

## Market Analysis

The market for medical simulation has grown significantly in the past year as the need for healthcare professionals to improve their skills in a safe environment has increased. In 2024, the global medical simulation market was valued at approximately USD 1.65 billion and is projected to grow at a compound annual growth rate (CAGR) of 11.21% through 2030.<sup>26</sup>



**Figure 4.1:** Graph of the projected medical simulation market from 2020 - 2030.

This expansion is fueled by a rising emphasis on patient safety, specifically from the increasing demand for minimally invasive surgeries, as well as technological advancements in simulation and a shortage of healthcare professionals. This simulation-based training in VR and AR can improve patient outcomes, and according to a WHO report, unsafe care is a major contributor to death and disability worldwide, with up to 134 million adverse events occurring in hospitals across low and middle-income countries, leading to approximately 3 million deaths annually. In high-income

<sup>26</sup>Medical Simulation Market Size, Share, Growth Report, 2030, [www.grandviewresearch.com/industry-analysis/medical-healthcare-simulation-market](http://www.grandviewresearch.com/industry-analysis/medical-healthcare-simulation-market). Accessed 24 Feb. 2025.

countries, about 1 in 10 patients experience an adverse event during hospital care, emphasizing the need for improved and safer healthcare practices worldwide.<sup>27</sup>

In the US, there are a growing number of medical residents and many graduates entering residency programs - a training environment designed to train graduates to become specialized doctors. As of the 2023-2024 academic year, there are 162,644 active residents and fellows, with 133,776 (82.3%) in specialty programs and 28,828 (17.7%) in subspecialty programs.<sup>28</sup> This increasing number of medical trainees shows the demand for more adjustable and accessible training solutions that can reinforce retention and performance in the healthcare workforce.

Our market would typically include medical students or other professionals aged 18-35 years old, and have a middle to high income or use institutional funding to buy our product. They would have a Bachelor’s degree or higher in education and be situated in areas with limited cadaver access or specialized simulation labs. Medical institutions already spend billions on training materials, so a budget-friendly alternative that can be reused and can minimize medical errors is attractive.

## Potential Profits and Distribution

The solution consists of 3 main components- the wearable glove frame, the force feedback system, and the electronics, each with its own sub-parts. The frame is made from a lightweight, stretchable spandex glove, ensuring adaptability and comfort for different hand sizes. The force feedback system is composed of servo motors, potentiometers, and elastic components to simulate resistance when touching or holding objects. The multitude of 3D-printed parts are integrated into the system to keep costs low, reduce weight, and allow for easy modifications or repairs. The estimated total cost to produce one unit is \$146.95, with an additional cost of about \$70 to print 3D parts. However, with bulk manufacturing and injection molding, per-unit costs can be further reduced to improve efficiency while maintaining affordability. Assuming an 8% sales tax and an average \$20 shipping fee, our wholesale price would be approximately \$260, and the suggested retail price will likely be \$400 to \$500, making this glove significantly less expensive than ~\$4000 competitors.

| Part Description                            | Total Price of Each |
|---------------------------------------------|---------------------|
| Unisex Adult 18" Spandex Full Finger Gloves | \$9.99              |
| MG90S servo with 360-degree rotation        | \$20.99             |

<sup>27</sup>Medical Simulation Market Size, Share, Growth Report, 2030, [www.grandviewresearch.com/industry-analysis/medical-healthcare-simulation-market](https://www.grandviewresearch.com/industry-analysis/medical-healthcare-simulation-market). Accessed 24 Feb. 2025.

<sup>28</sup>“ACGME Releases 2023-2024 Statistics on Graduate Medical Education Programs and Resident Physicians.” ACGME, [www.acgme.org/newsroom/2024/10/acgme-releases-2023-2024-statistics-on-graduate-medical-education-programs-and-resident-physicians/](https://www.acgme.org/newsroom/2024/10/acgme-releases-2023-2024-statistics-on-graduate-medical-education-programs-and-resident-physicians/). Accessed 24 Feb. 2025.

|                                                                    |         |
|--------------------------------------------------------------------|---------|
| Arduino Nano ESP32                                                 | \$22.00 |
| TWTADE 10Pcs Single Linear Rotary Seal Amplifier<br>Potentiometers | \$12.66 |
| JANYUN Black Retractable Badge Reels                               | \$12.99 |
| Mini Portable Charger 5000mAh, USB-C Power Bank                    | \$23.97 |
| USB 3.1 Type C 24 Male and Female Plug Socket Adapter              | \$7.49  |
| 20 AWG Electric Wire Flexible Silicone Wires                       | \$14.99 |
| 2 Rolls 32.2ft Straps 1 Inch Wide with 40 Metal Buckles            | \$9.98  |
| 48 x 3g Super Glue Clear                                           | \$11.89 |

**Figure 4.2:** Estimated cost breakdown of parts to outsource.

To market our product, we will reach out to universities and medical schools through demonstrations and partnerships, as well as provide an exhibit at VR technology conferences to showcase the effectiveness of our product. In addition, online advertising will be carried out through LinkedIn and other major social media platforms, and distribution will be handled through direct sales to institutions, online sales to customers, or retail partnerships where our haptic VR glove can be on display for testing. Lastly, the global medical simulation market is projected to exceed \$5 billion by 2026<sup>29</sup>, and with VR-based medical training growing significantly due to its ability to improve patient outcomes and reduce training costs, capturing just 0.5% of the market represents \$25 million.

## V. Conclusion

Overall, the VR glove offers a cost-effective, sustainable, and immersive solution for virtual medical training, reducing reliance on cadaver-based education whilst enhancing accessibility and precision. By integrating a physical glove (with force feedback, and tracking) with an open-source application (Unity-based simulation with compatibility with existing infrastructure), the glove holds up its efficiency and showcases itself as a competitive, marketable product considering both sustainability and usability.

<sup>29</sup>“Healthcare Simulation Market Growth, Drivers, and Opportunities.” *MarketsandMarkets*, [www.marketsandmarkets.com/Market-Reports/healthcare-medical-simulation-market-1156.html](https://www.marketsandmarkets.com/Market-Reports/healthcare-medical-simulation-market-1156.html). Accessed 24 Feb. 2025.